
EE 513: EE 513

Course Code	Special Electrical Machines
Course Name	EE 513
L-T-P-C	3-0-0-3
Prerequisites	EE 201 Electromechanics
Intended for	UG/PG
Distribution	Elective
Approval	8th Senate (Oct 2015)

Course Contents

Permanent Magnet Brushless D.C. Motors: Fundamental equations EMF and Torque equations Torque speed characteristics Rotor position sensing Sensorless motors Motion control. [9 Lectures]

Permanent Magnet Synchronous Motors: Construction - Principle of operation EMF and torque equations Starting Rotor configurations Dynamic model. [9 Lectures]

Synchronous Reluctance Motors: Constructional features axial and radial flux motors operating principle characteristics. [8 Lectures]

Switched Reluctance Motors: Constructional features principle of operation torque production characteristics power controllers. [8 Lectures]

Stepping Motors: Features fundamental equations PM stepping motors Reluctance stepping motors Hybrid stepping motors Torque and voltage equations characteristics. [8 Lectures]

Text Books

1. Miller, T. J. E., **Brushless Permanent Magnet and Reluctance Motor Drives**, Oxford Science Publications, 1989.
2. Kenjo, T., and Sugawara, A., **Stepping Motors and their Microprocessor Controls**, Oxford Science Publications, 1984.
3. Venkataratnam K., **Special Electrical Machines**, CRC Press, 2009.

References

1. Krishnan, R., **Permanent Magnet and BLDC Motor Drives**, CRC Press, 2009.
2. Chang-liang, X., **Permanent Magnet Brushless DC Motor Drives and Controls**, 2012.